

HISTOPATHOLOGICAL EVALUATION OF ENDOMETRIAL BIOPSIES AND CURETTAGE IN ABNORMAL UTERINE BLEEDING

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BACKGROUND & SIGNIFICANCE

What is Abnormal Uterine Bleeding (AUB)?

Abnormal uterine bleeding (AUB) is an umbrella term encompassing various forms of uterine bleeding that deviate from the normal menstrual pattern in terms of regularity, volume, duration, or frequency.

14-25%

Reproductive
age women
affected

50%

Perimenopausal
women
affected

#1

Reason for
gynecologic
consultation

AUB significantly impacts quality of life, causing anemia, fatigue, and activity restrictions. Early diagnosis is crucial for identifying underlying pathologies including hyperplasia and malignancy.

The PALM-COEIN system

- P – Polyp (AUB-P)
- A – Adenomyosis (AUB-A)
- L – Leiomyoma (AUB-L)
- M – Malignancy and Hyperplasia (AUB-M)
- C – Coagulopathy (AUB-C)
- O – Ovulatory Dysfunction (AUB-O)
- E – Endometrial (AUB-E)
- N – Iatrogenic and Not Otherwise Classified (AUB-N)

RESEARCH OBJECTIVES

Objective 1:

To identify the various histopathological patterns in the endometrial biopsies and curetting in the patients of AUB.

Objective 2:

To see the correlation of the endometrial Biopsies and curetting in AUB with the radiological findings (if any) in these patients.

Study Significance

This research aims to enhance diagnostic accuracy and improve clinical management of AUB through comprehensive histopathological analysis and clinical correlation.

METHODOLOGY

Sample Collection

- Dilatation and curettage under sedation
- 351 patients with AUB included
- Fixed in 10% neutral buffered formalin

Processing & Analysis

- 5 μ thickness sections prepared
- H&E staining performed
- WHO classification used for reporting

Clinical Data Collection

Age, complaints, obstetric history, menstrual history, drug history, and clinical diagnosis collected from histopathological requisition forms

Statistical Analysis

Data analyzed using SPSS Ver. 23 with appropriate statistical tests. P-value <0.05 considered statistically significant.

PATIENT DEMOGRAPHICS

351

Total Patients

47.6%

**Age 31-40
years**

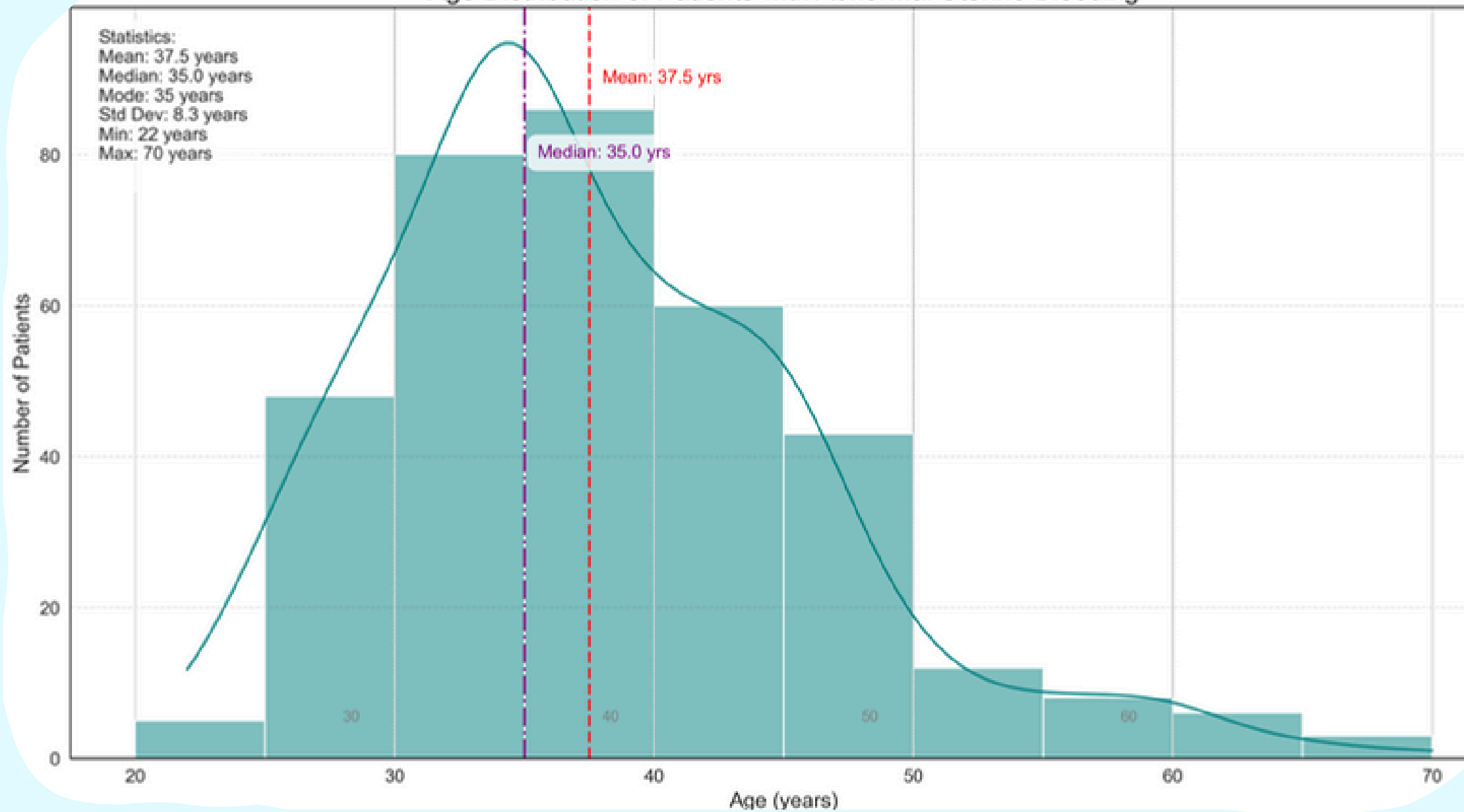
92.6%

Low parity

25.4%

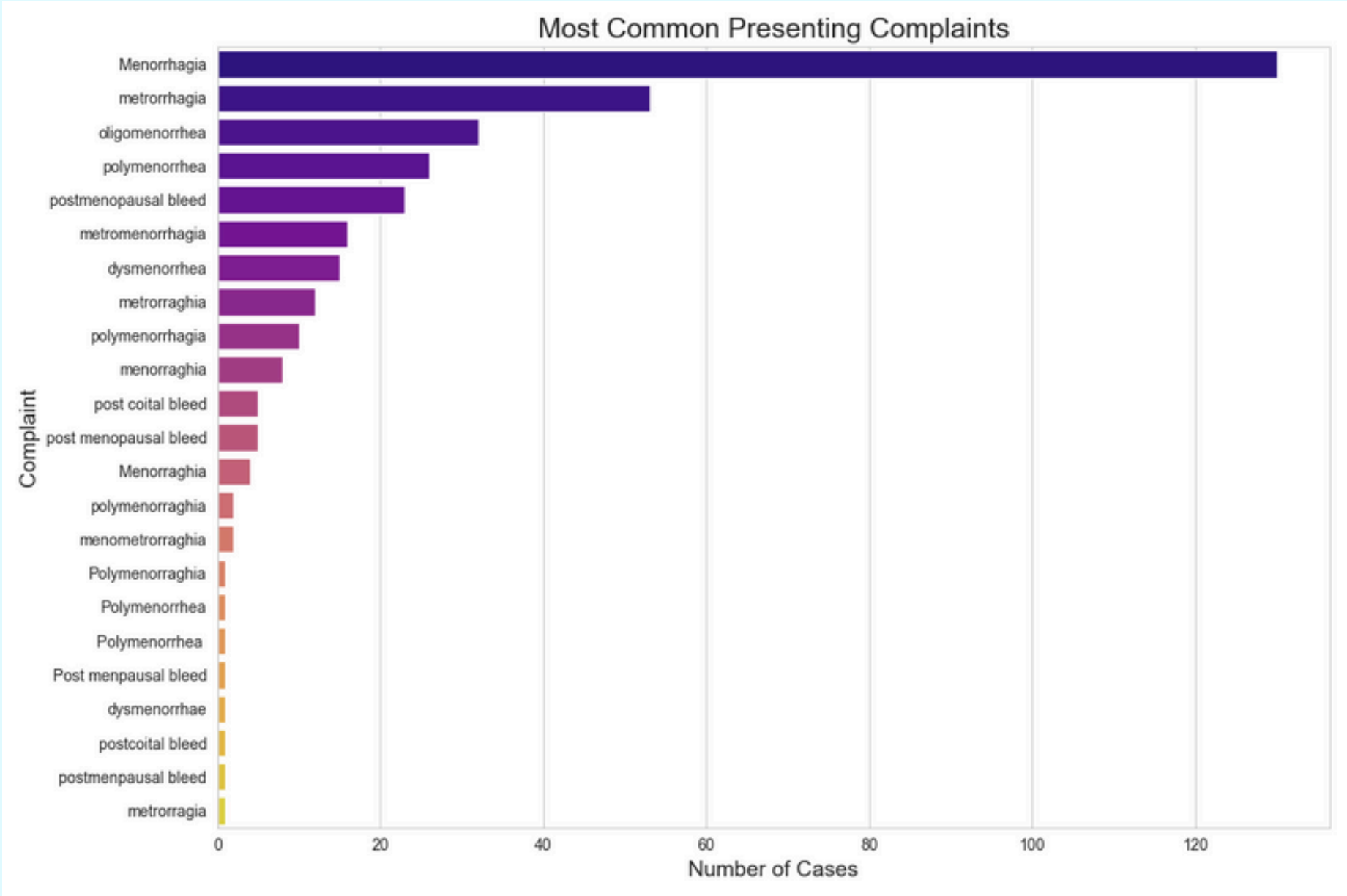
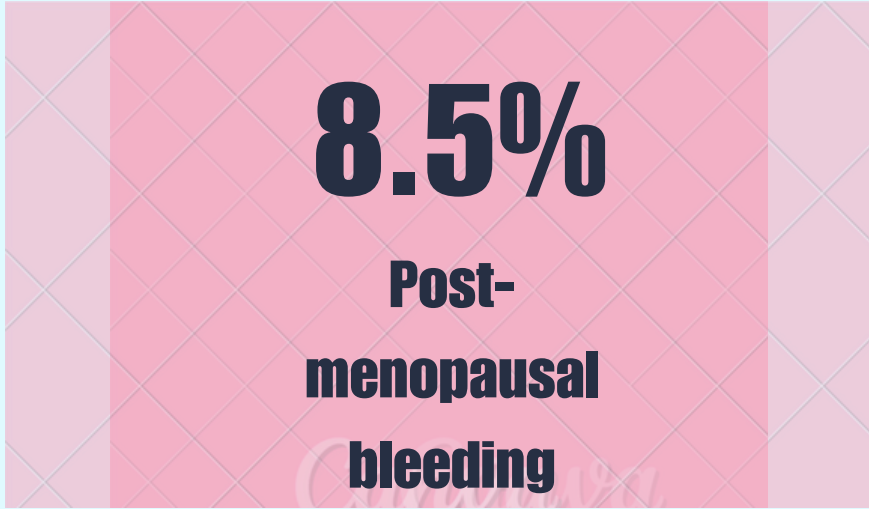
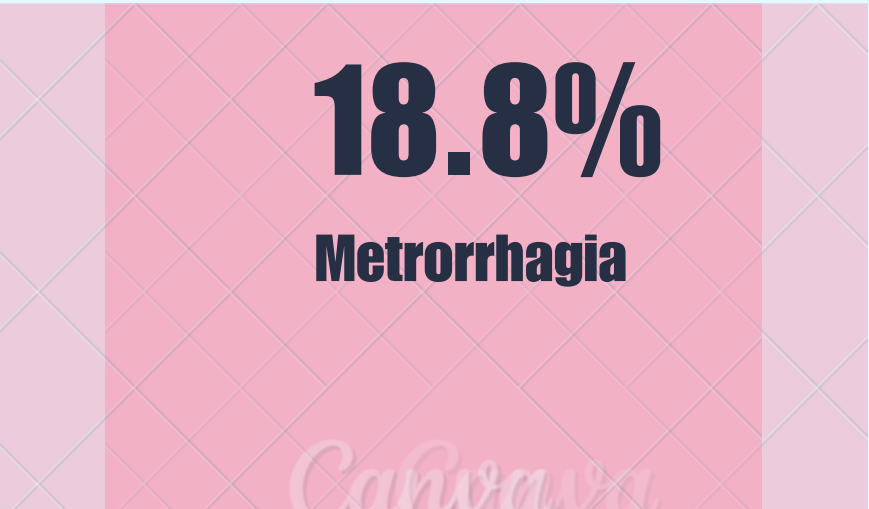
**Hormonal drug
history**

Age Distribution of Patients with Abnormal Uterine Bleeding



The majority of patients were in the reproductive and perimenopausal age groups, with most being of low parity and having no history of hormonal drug intake

CLINICAL PRESENTATION



Menorrhagia emerged as the most common presenting complaint, highlighting the prevalence of heavy bleeding patterns that prompt medical evaluation.

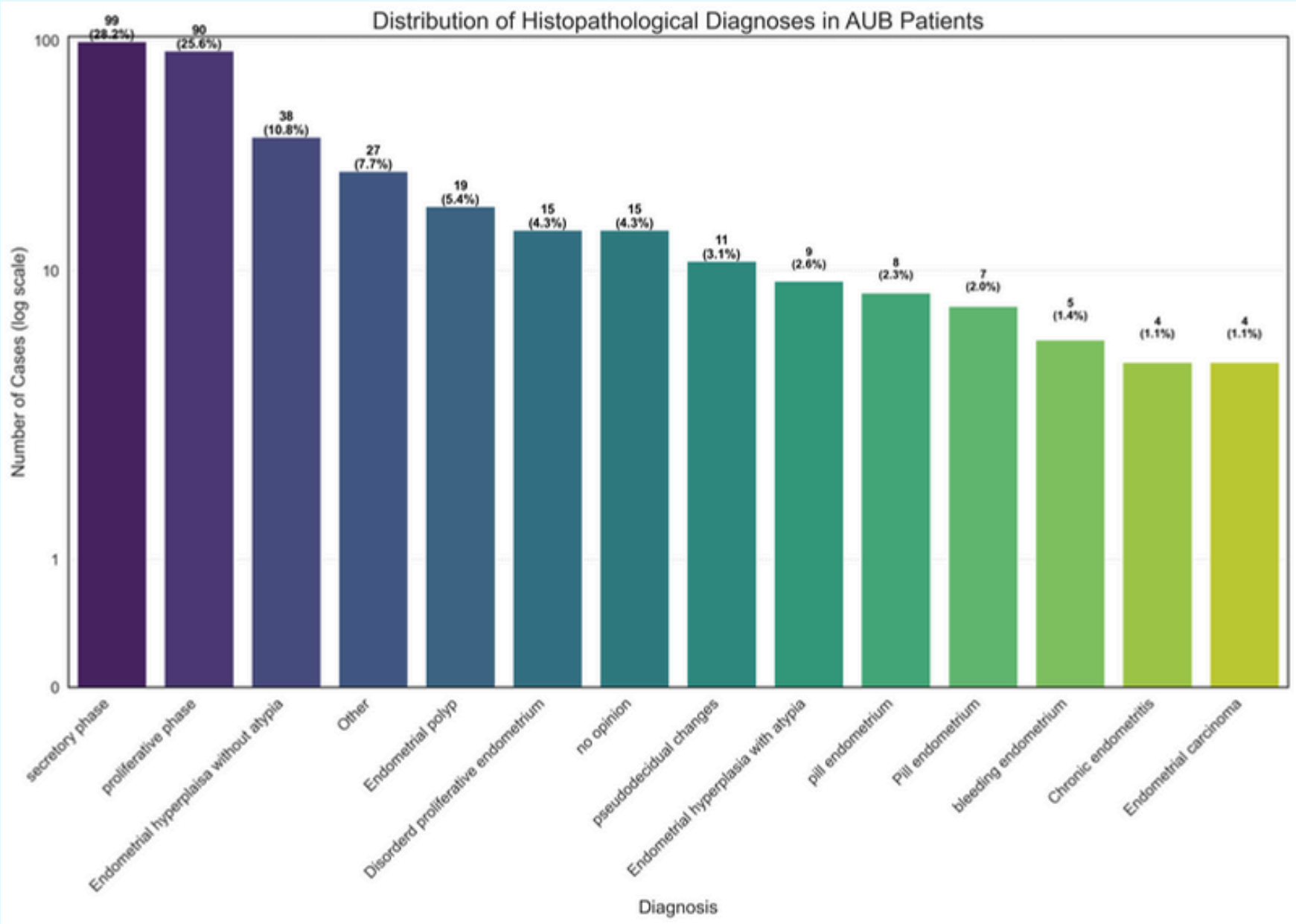
KEY HISTOPATHOLOGICAL FINDINGS

29.3%
**Secretory
Phase**

28.2%
**Proliferative
Phase**

12.0%
**Hyperplasia
(no atypia)**

2.8%
**Atypical
Hyperplasia**



Over 57% of cases showed normal cyclical changes, while significant pathological findings included hyperplasia and malignancy requiring immediate attention.

PATIENT DEMOGRAPHICS

Histopathology ↔ LMP

$r = 0.766, p < 0.001$

Strong positive correlation - menstrual timing closely aligns with histological patterns

Histopathology ↔ Drug History

$r = 0.287, p < 0.001$

Moderate correlation - hormonal intake influences endometrial morphology

Histopathology ↔ Age

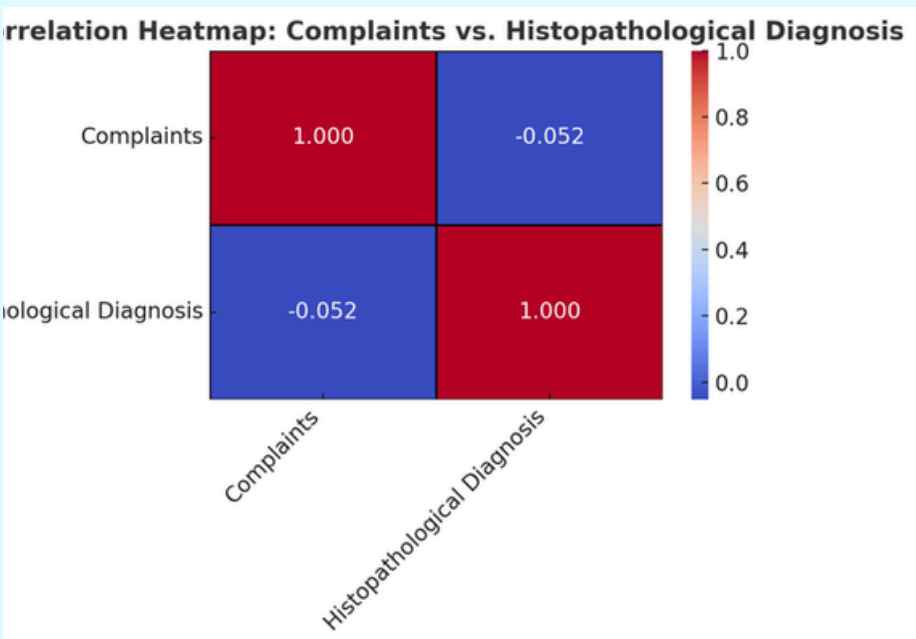
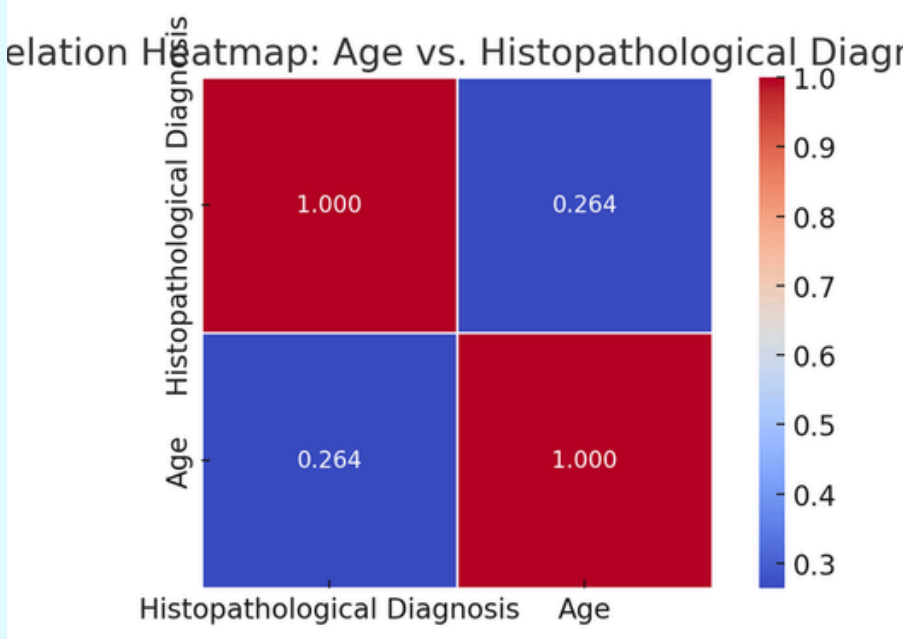
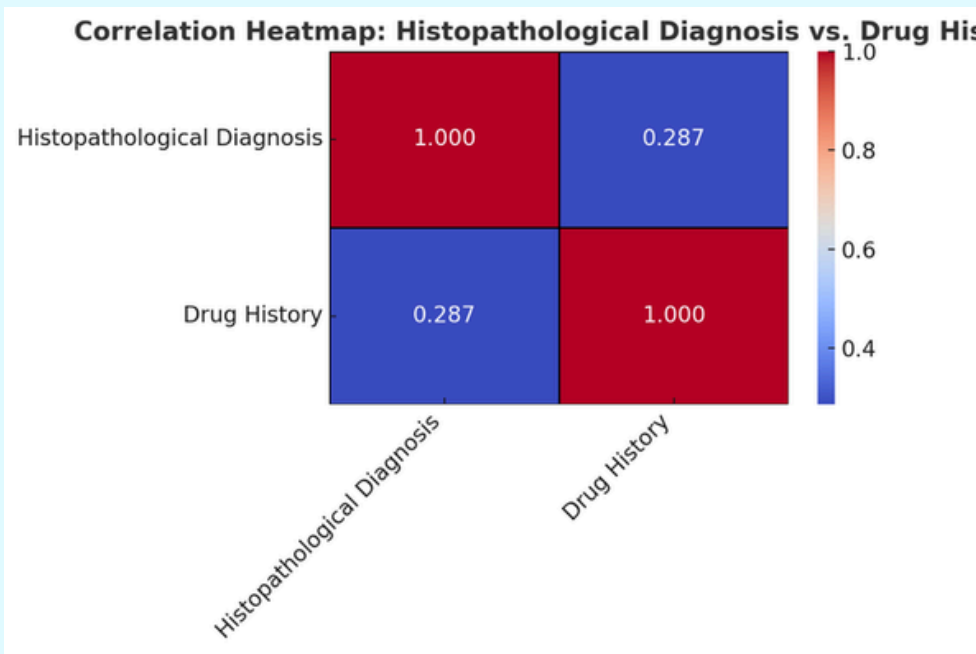
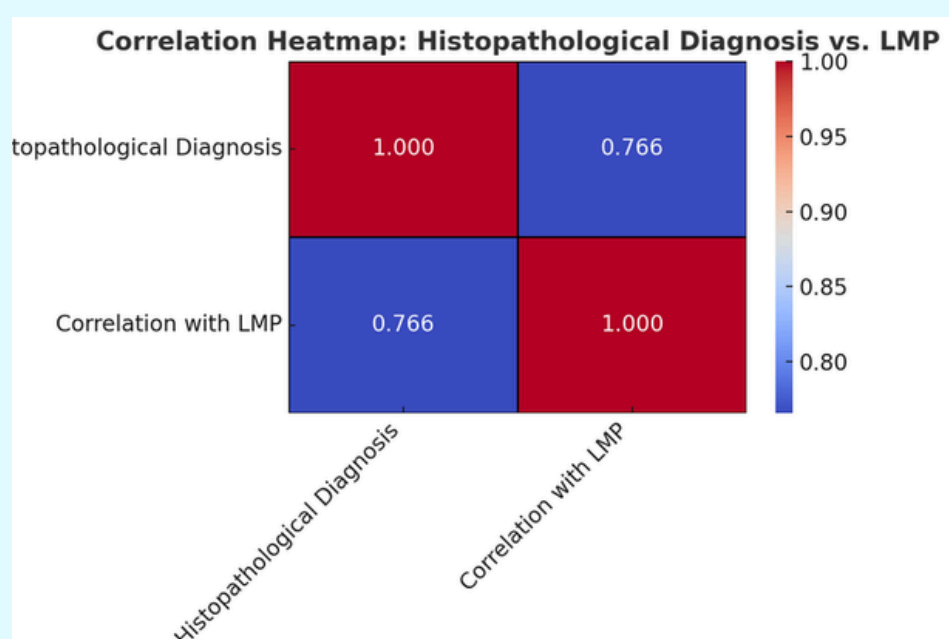
$r = 0.264, p < 0.001$

Moderate correlation - pathological changes increase with age

Histopathology ↔ Complaints

$r = -0.052, p = 0.331$

No significant correlation - symptoms don't predict pathology



CLINICAL IMPLICATIONS & RECOMMENDATIONS

- **Routine Histopathological Evaluation:** Essential for all AUB patients, especially those >35 years and perimenopausal women
- **Menstrual History Documentation:** LMP correlation emphasizes the critical role of menstrual history in interpretation
- **Hormonal Drug History:** Must be documented as it significantly influences endometrial morphology
- **Symptom Limitations:** Clinical complaints alone cannot predict underlying pathology
- **Age-Based Risk Stratification:** Increasing age correlates with more advanced pathological changes

CLINICAL IMPLICATIONS & RECOMMENDATIONS

Major Findings

While most AUB cases reflect benign cyclical changes, a significant proportion present with premalignant or malignant conditions not evident through clinical assessment alone.

Diagnostic Standard

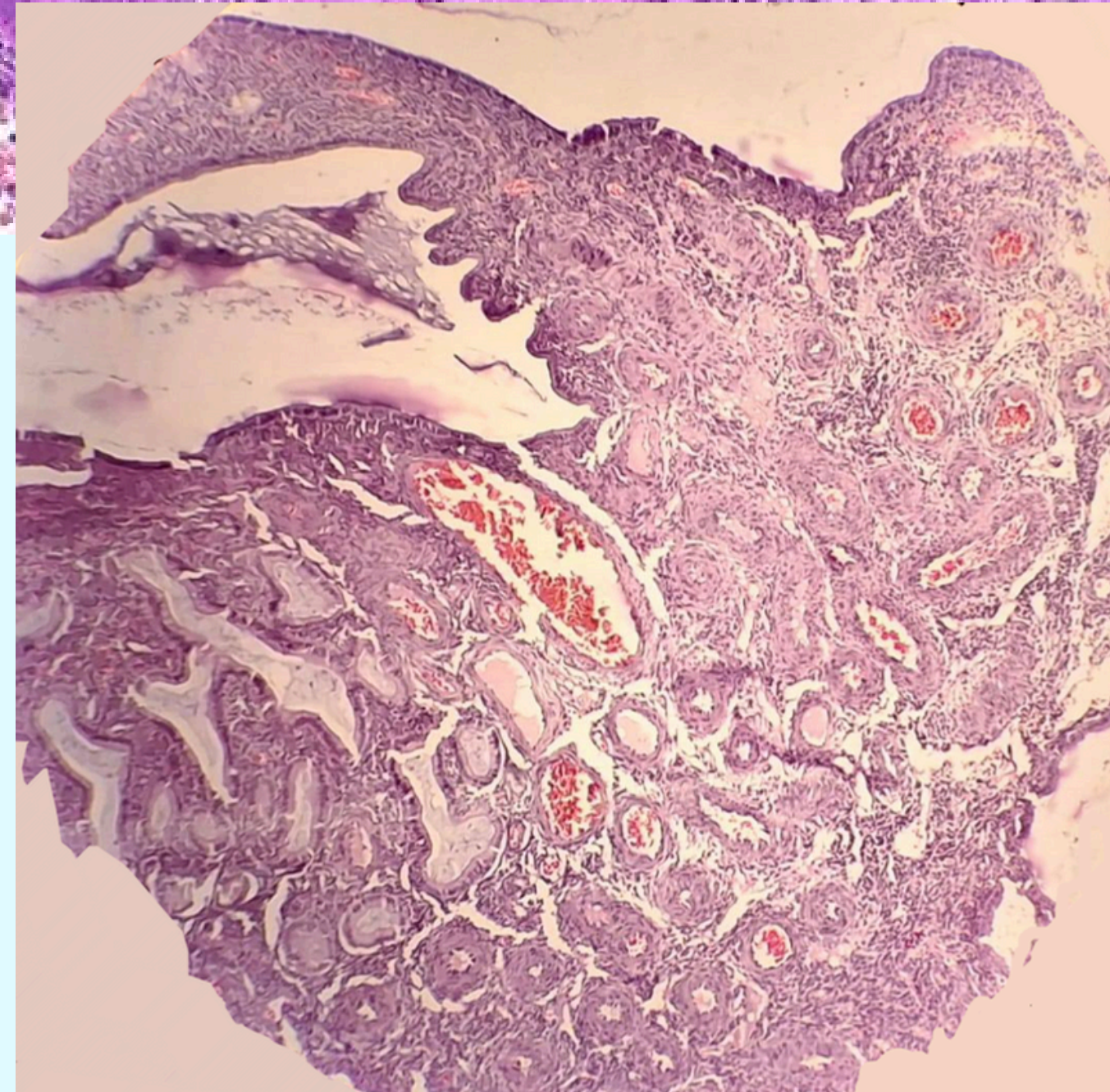
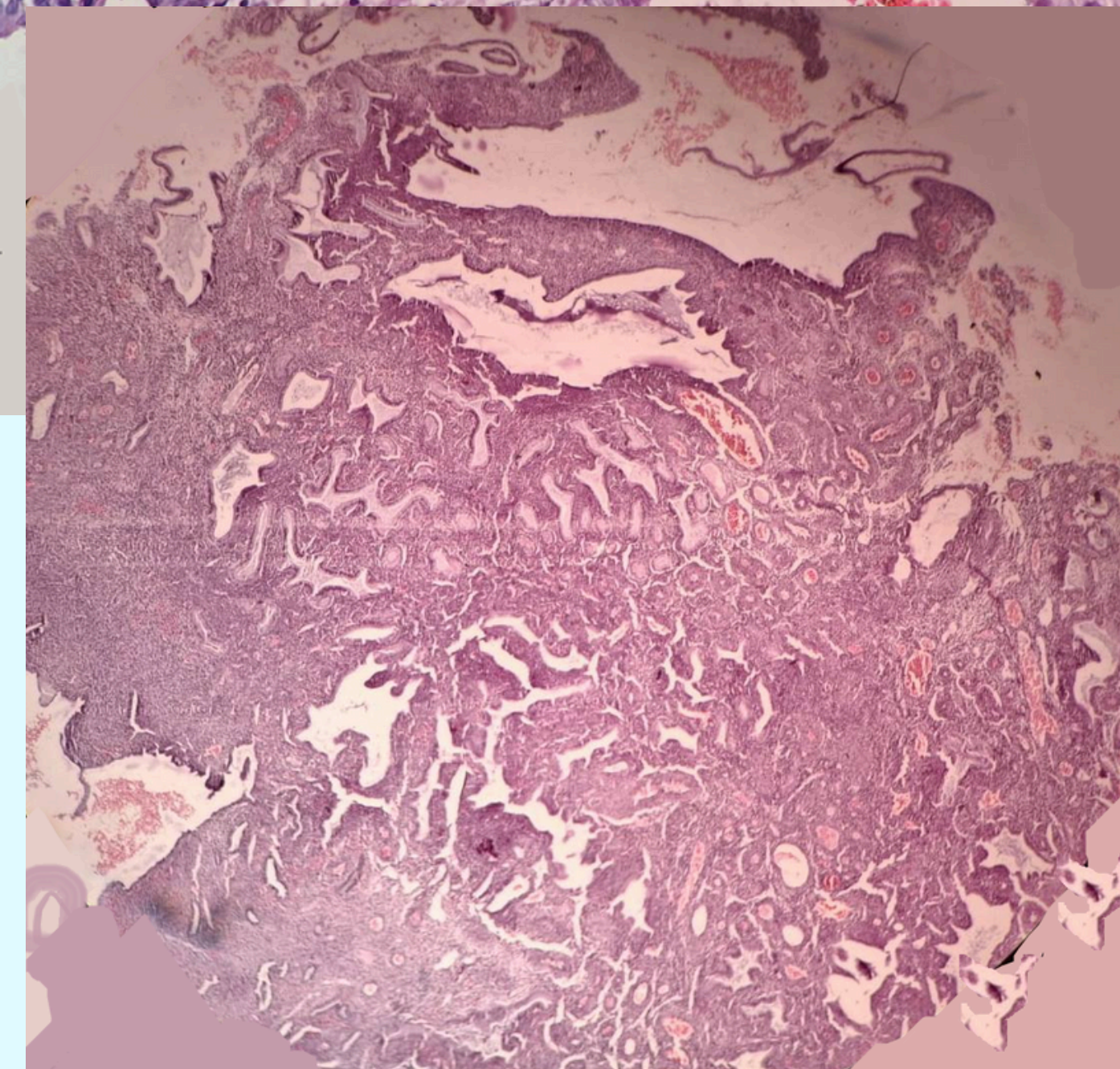
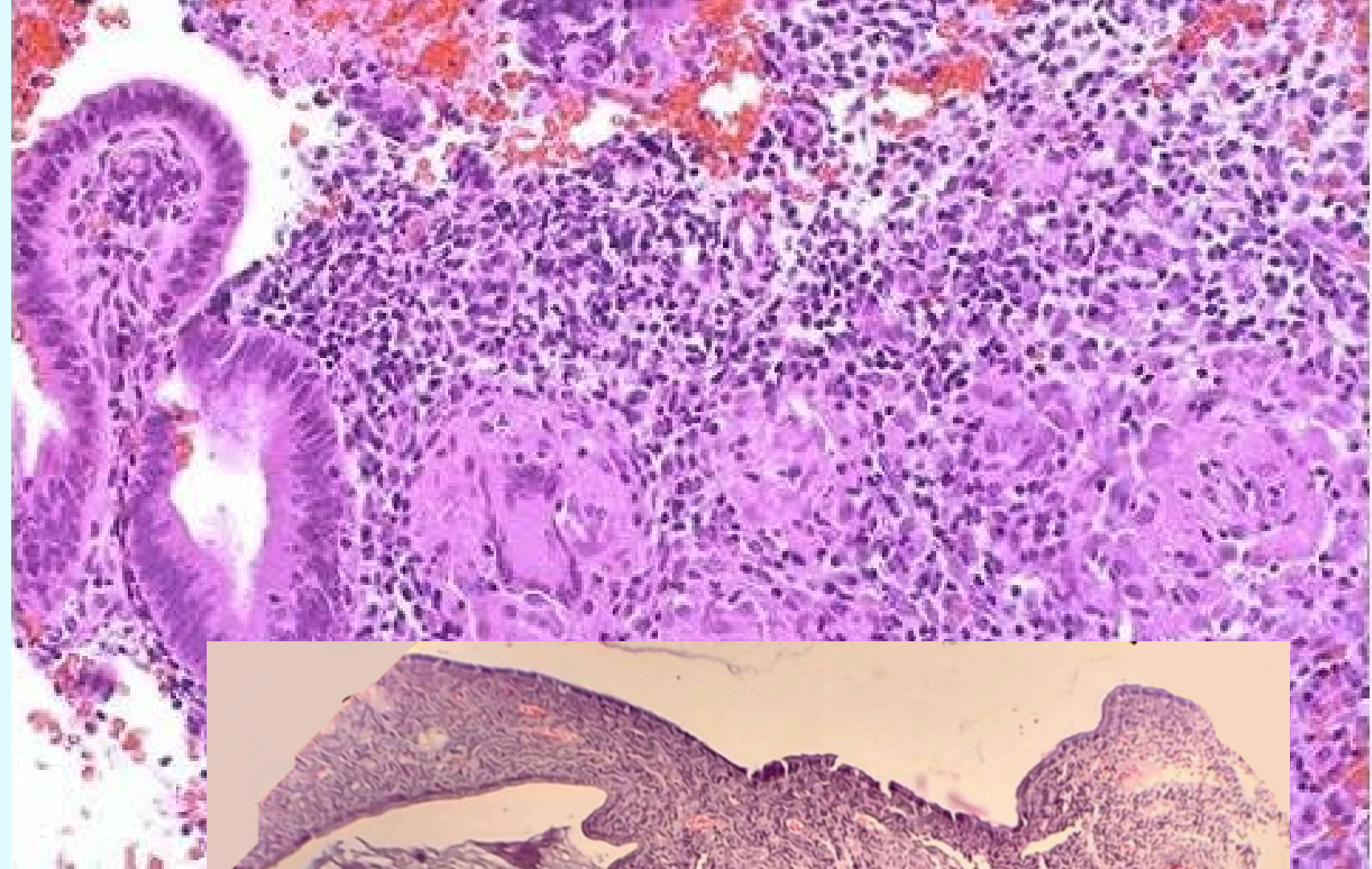
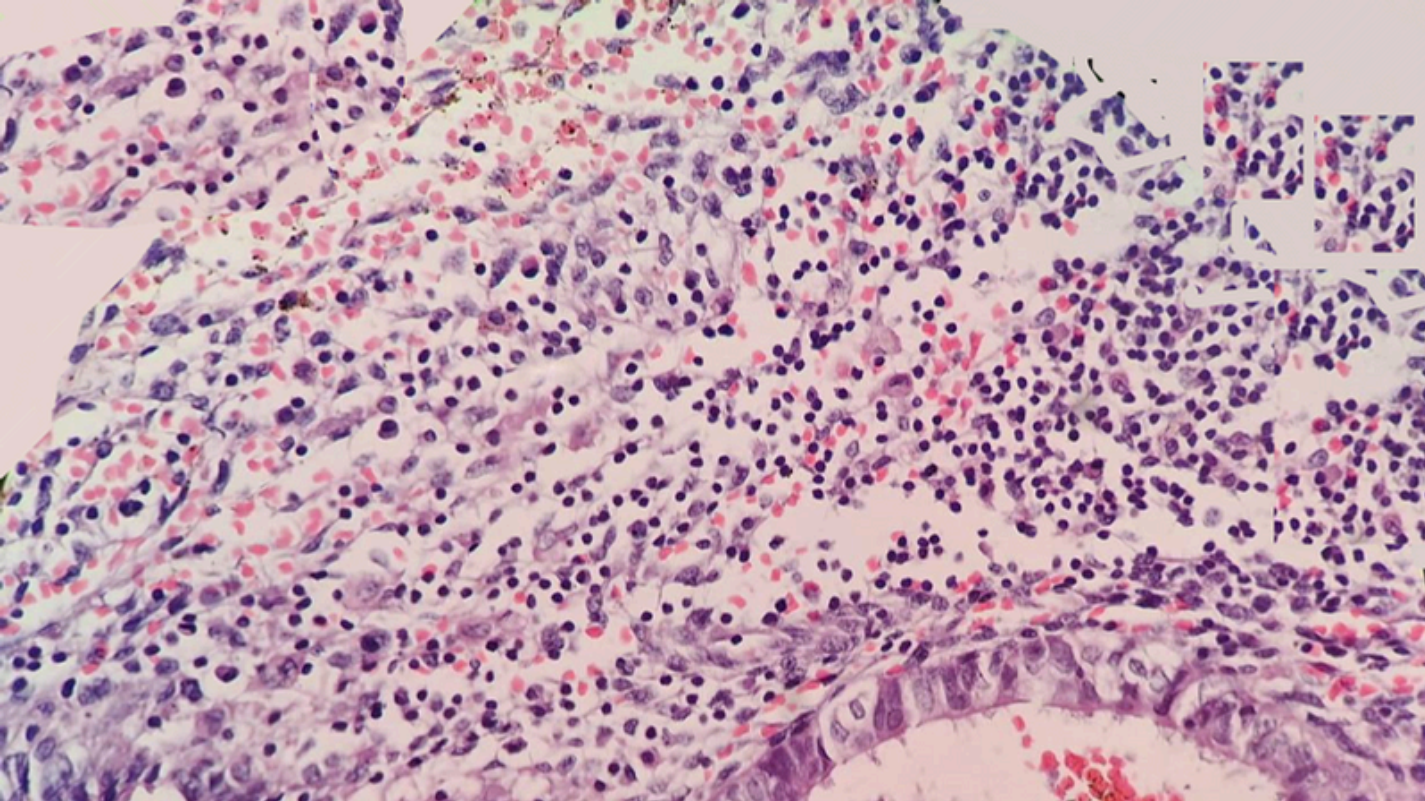
Histopathological examination remains the gold standard for accurately identifying underlying AUB pathology

Clinical Integration

Strong LMP correlation and moderate hormonal associations emphasize integrated diagnostic approaches

Future Direction

This study advocates for comprehensive, standardized AUB evaluation incorporating detailed clinical history, ultrasonographic findings, and routine histopathological sampling for optimal patient care.



**THANK
YOU**

